IOT PROJECT

Sleep-Awake Monitoring and Detection using Laptop Camera and Arduino

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Title:Sleep-Awake Monitoring and Detection using Laptop Camera and Arduino

1. Introduction

The increasing number of accidents caused by drowsiness or sleep, especially in drivers or machine operators, highlights the need for a smart monitoring system. This project uses Computer Vision (OpenCV) to detect the sleep state based on eye closure and sends signals to an Arduino-based alert system (motor, buzzer, LEDs) to take action.

2. Objective

To develop an IoT-based solution that monitors a person's eyes using a webcam and triggers alerts (buzzer, LED, motor) via Arduino when sleep or drowsiness is detected.

Components Required

Hardware:

- Arduino UNO/Nano

- Breadboard

- Motor (DC)

- L298N Motor Driver Module

- Buzzer

- Red & Green LEDs

- 220-ohm Resistors

- Jumper Wires

- USB Cable

Software:

- Arduino IDE

- Python 3.x

- OpenCV Library

- dlib Library

- pyserial Library

- shape_predictor_68_face_landmarks.dat (for facial landmarks)

Block Diagram

[Laptop Camera (OpenCV) + Python] --> [Serial Communication] --> [Arduino]

|

[Buzzer | Motor | Red & Green LEDs]

3. Working Principle

1. Python and OpenCV capture real-time video from the laptop camera.

2. Using facial landmarks, the system calculates Eye Aspect Ratio (EAR).

3. If EAR drops below a threshold for a certain number of frames, it means the eyes are closed → sleep detected.

4. A signal 'S' is sent to Arduino to activate the buzzer, red LED, and motor.

5. If eyes are open (awake), signal 'A' is sent and green LED glows.

Methodology

This project uses a combination of **Computer Vision (OpenCV)** and **Arduino microcontroller** to implement a real-time sleep-awake detection system.

1. Software (Python with OpenCV)

- **Step 1: Video Capture**
    - The laptop webcam is accessed using OpenCV to continuously capture video frames.
- **Step 2: Eye Detection using Haar Cascades**
    - Pre-trained Haar Cascade classifiers (provided by OpenCV) are used to detect the face and eyes in each frame.

    - Once eyes are detected, their presence or absence in consecutive frames is analyzed.

- **Step 3: Sleep Detection Logic**
    - If eyes are **not detected** for a specific number of continuous frames (e.g., 20 frames), it is assumed the person is sleeping.
    - If eyes are detected consistently, the system determines the user is awake.
- **Step 4: Serial Communication**
    - Depending on the detection:
        - 'S' is sent over serial to the Arduino if the person is likely asleep.
        - 'A' is sent if the person is awake.

2. Hardware (Arduino with Sensors and Actuators)

- **Step 1: Serial Data Handling**
    - Arduino continuously reads data from the serial port sent by the Python program.
- **Step 2: Control Outputs**
    - If 'S' is received:
        - **Red LED** turns ON to signal sleep.
        - **Buzzer** sounds an alert.
        - **Motor** vibrates for haptic feedback.
    - If 'A' is received:
        - **Green LED** turns ON indicating the person is awake.
        - Buzzer and motor remain OFF.

This approach enables effective monitoring of a user's eye activity using only OpenCV, eliminating the need for heavy machine learning models or deep learning libraries, making it lightweight and efficient.

4.Procedure-implementation

Hardware Setup:

1. **Connect the Arduino:**
    - Place the Arduino board on a breadboard.
    - Connect the L298N motor driver to the Arduino:
        - IN1, IN2 to Arduino digital pins (e.g., D7, D8)
        - ENA to 5V or use PWM pin (for speed control)
        - Connect the motor to OUT1 and OUT2 of L298N
        - Connect power and GND appropriately
2. **Buzzer and LEDs:**
    - Connect the **buzzer** to a digital pin (e.g., D9) with a resistor.
    - Connect **red LED** to a digital pin (e.g., D10), and **green LED** to another (e.g., D11).
    - Use 220-ohm resistors in series with each LED.
3. **Power Supply:**
    - Power the Arduino via USB from the laptop.
    - Power the motor driver and motor with a separate battery (if needed).

Software Setup:

Python Script (on Laptop):

1. Install Python and required libraries:
2. pip install opencv-python dlib imutils pyserial
3. Download the shape_predictor_68_face_landmarks.dat file.
4. Write the Python script to:
    - Capture webcam video.
    - Use dlib to detect face and eyes.
    - Calculate Eye Aspect Ratio (EAR).

- Send 'S' for sleep or 'A' for awake over serial to Arduino.

Arduino Sketch:

1. Open Arduino IDE and write a script to:
    - Read serial input from Python.
    - Turn ON/OFF motor, LEDs, and buzzer based on signal.
2. Upload the code to Arduino via USB.

Execution:

1. Run the **Python script** to start video capture and eye monitoring.
2. If the person closes their eyes beyond the threshold time:
    - Python sends 'S' to Arduino.
    - Arduino activates **red LED**, **buzzer**, and **motor**.
3. When the person's eyes are open:
    - Python sends 'A'.
    - Arduino turns ON **green LED** and disables alerts.

Code Explanation

Python: Captures video feed, detects eyes, sends sleep/awake signal to Arduino.

```
import cv2

import numpy as np

import serial

import time

import logging
```

```
# Setup logging
logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
logger = logging.getLogger(_name_)

def initialize_serial(port='COM4', baud=9600):
    """Initialize serial communication."""
    try:
        ser = serial.Serial(port, baud, timeout=1)
        time.sleep(2)  # Wait for connection
        logger.info(f"Connected to {port}")
        return ser
    except serial.SerialException as e:
        logger.error(f"Serial error: {e}")
        return None

def initialize_camera(index=0):
    """Initialize webcam."""
    cap = cv2.VideoCapture(index)
    if not cap.isOpened():
        logger.error("Failed to open webcam")
        return None
    logger.info("Webcam initialized")
    return cap

def load_cascades():
```

```
    """Load Haar cascades."""
    face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades +
'haarcascade_frontalface_default.xml')
    eye_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_eye.xml')
    if face_cascade.empty() or eye_cascade.empty():
        logger.error("Failed to load cascades")
        return None, None
    logger.info("Cascades loaded successfully")
    return face_cascade, eye_cascade


def detect_eyes(frame, face_cascade, eye_cascade):
    """Detect eyes in frame."""
    try:
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        faces = face_cascade.detectMultiScale(gray, scaleFactor=1.2, minNeighbors=5,
minSize=(30, 30))

        eyes_detected = False
        for (x, y, w, h) in faces:
            roi_gray = gray[y:y+h, x:x+w]
            roi_color = frame[y:y+h, x:x+w]
            eyes = eye_cascade.detectMultiScale(roi_gray, scaleFactor=1.1, minNeighbors=10,
minSize=(20, 20))

            if len(eyes) >= 1:
                eyes_detected = True
                for (ex, ey, ew, eh) in eyes:
```

```
                cv2.rectangle(roi_color, (ex, ey), (ex+ew, ey+eh), (0, 255, 0), 2)
            cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0, 0), 2)

        return eyes_detected, frame
    except Exception as e:
        logger.error(f"Eye detection error: {e}")
        return False, frame

def main():
    """Main function to run the eye-tracking IoT system."""
    # Initialize components
    ser = initialize_serial('COM4')
    cap = initialize_camera()
    face_cascade, eye_cascade = load_cascades()

    if ser is None or cap is None or face_cascade is None:
        logger.error("Initialization failed. Exiting...")
        return

    try:
        while True:
            ret, frame = cap.read()
            if not ret:
                logger.warning("Failed to capture frame")
                continue
```

```
            # Detect eyes
            eyes_open, frame = detect_eyes(frame, face_cascade, eye_cascade
            # Send command to Arduino
            command = b'1' if eyes_open else b'0'
            try:
                ser.write(command)
                logger.info(f"Sent: {command.decode()} ({'Eyes open' if eyes_open else 'Eyes closed'})")
            except serial.SerialException as e:
                logger.error(f"Serial write error: {e}")
            # Display frame
            cv2.imshow('Eye Detection', frame
            # Exit on 'q'
            if cv2.waitKey(1) & 0xFF == ord('q'):
                break
    except KeyboardInterrupt:
        logger.info("Program interrupted by user")
    except Exception as e:
        logger.error(f"Main loop error: {e}")
    finally:
        if cap:
            cap.release()
        if ser:
            ser.close()
        cv2.destroyAllWindows()
        logger.info("Resources cleaned up")
```

```
if _name_ == "_main_":
   main()
```

****Arduino:** Reads signal and activates buzzer, LEDs, and motor accordingly.**

```
// Pin definitions
const int buzzerPin = 9;    // Buzzer
const int motorPin = 10;    // DC motor (via driver)
const int redLedPin = 11;   // Red LED
const int greenLedPin = 12; // Green LED

void setup() {
  // Initialize pins
  pinMode(buzzerPin, OUTPUT);
  pinMode(motorPin, OUTPUT);
  pinMode(redLedPin, OUTPUT);
  pinMode(greenLedPin, OUTPUT);

  // Set initial state
  digitalWrite(buzzerPin, LOW);
  digitalWrite(motorPin, LOW);
  digitalWrite(redLedPin, LOW);
  digitalWrite(greenLedPin, LOW);

  // Start serial
  Serial.begin(9600);
  Serial.println("Arduino Ready"); // Debug message
}
```

```
void loop() {
  if (Serial.available() > 0) {
    char state = Serial.read();
    Serial.print("Received: "); // Debug
    Serial.println(state);

    if (state == '0') { // Eyes closed
      digitalWrite(buzzerPin, HIGH);
      digitalWrite(motorPin, LOW);
      digitalWrite(redLedPin, HIGH);
      digitalWrite(greenLedPin, LOW);
      Serial.println("Eyes closed: Buzzer ON, Motor OFF, Red ON");
    } else if (state == '1') { // Eyes open
      digitalWrite(buzzerPin, LOW);
      digitalWrite(motorPin, HIGH);
      digitalWrite(redLedPin, LOW);
      digitalWrite(greenLedPin, HIGH);
      Serial.println("Eyes open: Buzzer OFF, Motor ON, Green ON");
    } else {
      Serial.println("Invalid command");
    }
  }
}
```

5.Oservations:

- Real-time Eye Detection Accuracy

- The system was able to consistently detect the user's eye status (open/closed) using a standard laptop webcam.
- Eye Aspect Ratio (EAR) threshold and frame count were effective in avoiding false triggers (e.g., blinking was not misclassified as sleep).

 Reliable Communication with Arduino

- Serial communication between the Python script and Arduino was stable.
- Commands ('S' for sleep, 'A' for awake) were successfully received by Arduino without lag.

 Effective Hardware Response

- Upon detection of a sleep state, the buzzer, red LED, and motor were activated reliably.
- In awake state, green LED turned on and all alerts were turned off smoothly.

 Environmental Factors

- Lighting played a role in detection accuracy. Proper lighting improved face and eye detection performance.
- Detection was slightly less responsive when the face was at an angle or partially out of the camera frame.

 Low Resource Usage

- The system worked efficiently on a basic laptop without requiring a GPU or external processing power.
- Arduino controlled components consumed very low power, making the project suitable for portable or vehicle-based use.

 Scalability

- The system is scalable and can be extended with additional sensors (like heartbeat, motion, etc.) or IoT-based dashboards (e.g., Blynk).

Applications

- Driver Drowsiness Detection

- Operator Monitoring in Factories

- Smart Health Monitoring Systems

- Surveillance and Security Systems

Advantages

- Real-time detection using a normal webcam

- Easy integration with alert systems using Arduino

- Portable and low-cost setup

Future Scope

- Add cloud connectivity using Blynk/IoT dashboard

- Use a pulse sensor or brainwave sensor for additional accuracy

- Integrate voice alerts or mobile notifications

result:

Sleep-Awake Monitoring and Detection System was successfully implemented using a **laptop camera with OpenCV and Python** for real-time eye monitoring and an **Arduino-based alert system** for physical responses. The system effectively:

- Detected eye closure and drowsiness with high accuracy using Eye Aspect Ratio (EAR).
- Differentiated between blinking and actual sleep to reduce false alarms.
- Sent appropriate signals to the Arduino to trigger the buzzer, LED indicators, and motor for immediate alert.
- Worked efficiently in real-time with minimal delay and hardware resources.

Thus, the objective of designing a low-cost, real-time drowsiness detection system was achieved.